FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

PHYSICS

CLASS XII

WAVE OPTICS

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1. Monochromatic light of wavelength 588 nm is incident from air to water interface. Find the wave length and speed of the refracted light. The refractive index of water is 4/3.
2. Two plane monochromatic waves propagating in the same direction with amplitudes A and 2 A and differing in phase by π/3 rad superpose. Calculate the amplitude of the resultant wave.
3. Two sources of intensity I and 4 I are used in an interference experiment. Find the intensity at points where the waves from two sources superimpose with a phase difference (i) zero (ii) π/2 and (iii) π.
4. In a Young's double slit experiment, the intensity of light at a point on the screen where the path difference is λ is k units. Find the intensity at a point where the path difference is (i) λ/4 (ii) λ/3 and (iii) λ/2
5. Two coherent light waves of intensity 5×10^{-2} Wm^{-2} each superimpose and produce the interference pattern on a screen. At a point where the path difference between the waves is λ/6, λ being the wave length of the wave, find the
   (a) phase difference between the waves
   (b) resultant intensity at the point
   (c) resultant intensity in terms of the intensity at the maximum.
6. Laser light of wavelength 630 nm incident on a pair of slits produces an interference pattern in which the bright fringes are separated by 8.1 mm. A second light produces an interference pattern in which the fringes are separated by 7.2 mm. Calculate the wavelength of the second light.
7. In a Young's double slit experiment, fringes are obtained on a screen placed a certain distance away from the slits. If the screen is moved by 5 cm towards the slits, the fringe width changes by 30 μm. Given that the slits are 1 mm apart, calculate the wavelength of the light used.
8. A narrow slit is illuminated by a parallel beam of monochromatic light of wavelength λ equal to 600 A^0 and the angular width of the central maxima in the resulting diffraction pattern is measured. When the slit is next illuminated by light of wavelength λ', the angular width decreases by 30 %. Calculate the value of wavelength λ'.
9. Two wavelengths of sodium light 590 nm, 596 nm are used, in turn, to study the diffraction taking place at a single slit of aperture 2×10^{-4} m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the positions of first maximum of the diffraction pattern obtained in the two cases.
10. Estimate the angular separation between first order maximum and third order minimum of the diffraction pattern due to a single slit of width 1 mm, when light of wavelength 600 nm is incident normal to it.
11. Two sources of intensity I_1 and I_2 undergo interference in Young's double slit experiment. Show that $\frac{\text{Im}\, ax}{\text{Im}\, in} = \left(\frac{a_1+a_2}{a_1-a_2}\right)^2$, where a_1 and a_2 are the amplitudes of disturbance for two sources S_1 and S_2.

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